



[REFERENCE IMPLEMENTATION]

CLINICAL RISK PREDICTION · WEARABLE ACTIGRAPHY · PORTFOLIO

Postoperative Delirium Risk Prediction

A pre-op risk model combining clinical history with wearable sleep data — built as a research reference implementation, and reported with the limits a real clinical tool would need to clear first.

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Gradient-boosted risk model · actigraphy ablation study · SHAP-explained pre-op huddle tool

EXECUTIVE SUMMARY

A wearable-derived signal that helps — and a screening tool that still misses most cases

This project predicts postoperative delirium risk from pre-op clinical history plus a week of wearable actigraphy, in a synthetic 4,200-episode cohort. Adding actigraphy measurably improves discrimination — and the resulting screening tool, like most such tools, still misses more cases than it catches.



0.700

Cross-validated AUC-ROC —
modest, and honestly reported



+8.7%

Relative AUC lift from adding
actigraphy — clears the 8% target



50%

Precision at the top-20% flag —
half of alerts are false positives



38%

Sensitivity at the operating
threshold — most cases are missed



Bottom line

This is a screening triage aid, not a diagnostic tool — it is built and reported as a research reference implementation, not a validated clinical device, and every limitation below is stated for that reason.

Presentation agenda



01

The Clinical Problem

Why postoperative delirium is worth screening for



02

Data & Cohort Profile

4,200 episodes, a real frailty and ASA-class gradient



03

Feature Engineering

What the model sees — and a deliberate exclusion



04

Modeling: Gradient-Boosted Risk

A LightGBM classifier over tabular and wearable features



05

Evaluation & Validation

An honest look at discrimination, lift, and miss rate



06

Clinical Impact

What a risk flag means for a pre-op huddle, in practice



07

Deployment & Responsible Use

Dashboard, model card, and clinical-AI guardrails



SECTION ONE

The Clinical Problem

Delirium after surgery is common, under-recognized, and often preventable — if the highest-risk patients can be found before the operating room.

THE STAKES

A common, costly, and frequently missed complication

Postoperative delirium is an acute, fluctuating disturbance in attention and awareness following surgery — disproportionately affecting older, frailer patients, and frequently under-recognized at the bedside.



A real base rate

26.2% incidence in this cohort — POD is common enough that a blanket "treat everyone the same" pre-op pathway leaves real risk unaddressed.



Often preventable, rarely predicted

Established non-pharmacological bundles reduce POD incidence — but they work best when aimed at the patients most likely to need them.



A high, but informative, bar

0.6442 AUC-ROC using clinical history alone — the baseline this deck's actigraphy signal is measured against.

The question this project asks is narrow and deliberate: can a pre-op risk score help direct a known-safe prevention bundle to the patients most likely to need it?

Informal judgment, applied unevenly

Without a structured score, pre-op delirium-risk screening tends to fall to individual clinician judgment — valuable, but inconsistent across a busy surgical schedule.



No structured trigger

Whether a patient gets flagged for extra attention depends on which clinician sees the chart, and how busy the clinic is that day



Factors, not a score

Age and frailty are known risk factors — but combining them consistently, patient after patient, is exactly what a model is good at and humans are not



A signal that goes unused

Actigraphy and sleep data, where collected at all, rarely make it into the pre-op risk conversation in any structured way



The gap this project targets

A consistent, auditable score — not a replacement for clinical judgment, but a structured input into it

26 . 2%

POD incidence, this cohort

More than one in four episodes in this dataset ends in a POD diagnosis — a base rate high enough that a structured score has real room to add value over judgment alone.

A triage score, not a diagnosis

This model never diagnoses delirium — delirium is diagnosed at the bedside, after surgery, by a clinician. What it does is estimate a pre-op probability, to help direct a known-safe prevention bundle before surgery even happens.

What the model outputs

$P(\text{delirium} \mid \text{pre-op history} + \text{actigraphy})$

A single probability, computed before the operating room, from data already available at the pre-op visit.

What happens with that number



Above threshold: flagged for the Hospital Elder Life Program bundle — established, non-pharmacological, low-risk



Below threshold: standard care pathway, no bundle assigned



Always reviewed by a clinician at the pre-op huddle — never an autonomous decision

What this project set out to prove



Build a credible pre-op risk score

Combine established clinical risk factors into a single, reproducible probability.



Test whether wearables add signal

Quantify the discrimination gained by adding a week of pre-op actigraphy to clinical history alone.



Avoid an unnecessary sensitive feature

Show the model performs without using sex as an input, and confirm that choice against the data.



Report performance honestly

State AUC, precision, and sensitivity together — not just whichever number looks best alone.



Validate with proper cross-validation

Report fold-level AUC, not a single lucky train/test split.



Scope this as research, explicitly

Package a model card and guardrails that keep this a reference implementation, not a deployed device.



SECTION TWO

Data & Cohort Profile

4,200 surgical episodes, a real frailty and ASA-class risk gradient, and a week of pre-op wearable sleep data.

DATASET

One cohort, two data modalities

Every episode pairs a standard pre-op clinical chart with a week of wearable actigraphy — the exact combination the ablation study in Section 05 tests for added value.

Source files

pod_episodes.csv	4,200 episodes — clinical + actigraphy features	Cohort
actigraphy_sample_traces.npz	raw 7-day traces, 250-episode demo sample	Wearable
pod_lgbm.joblib	trained LightGBM model + threshold	Model
oof_predictions.csv	out-of-fold predictions, all 4,200 episodes	Eval
metrics.json	cross-validated AUC, precision, incidence	Eval
ablation_report.json	tabular-only vs. +actigraphy comparison	Eval

Cohort at a glance

4,200

surgical episodes

26.2%

POD incidence

13

features used by the model

5

cross-validation folds

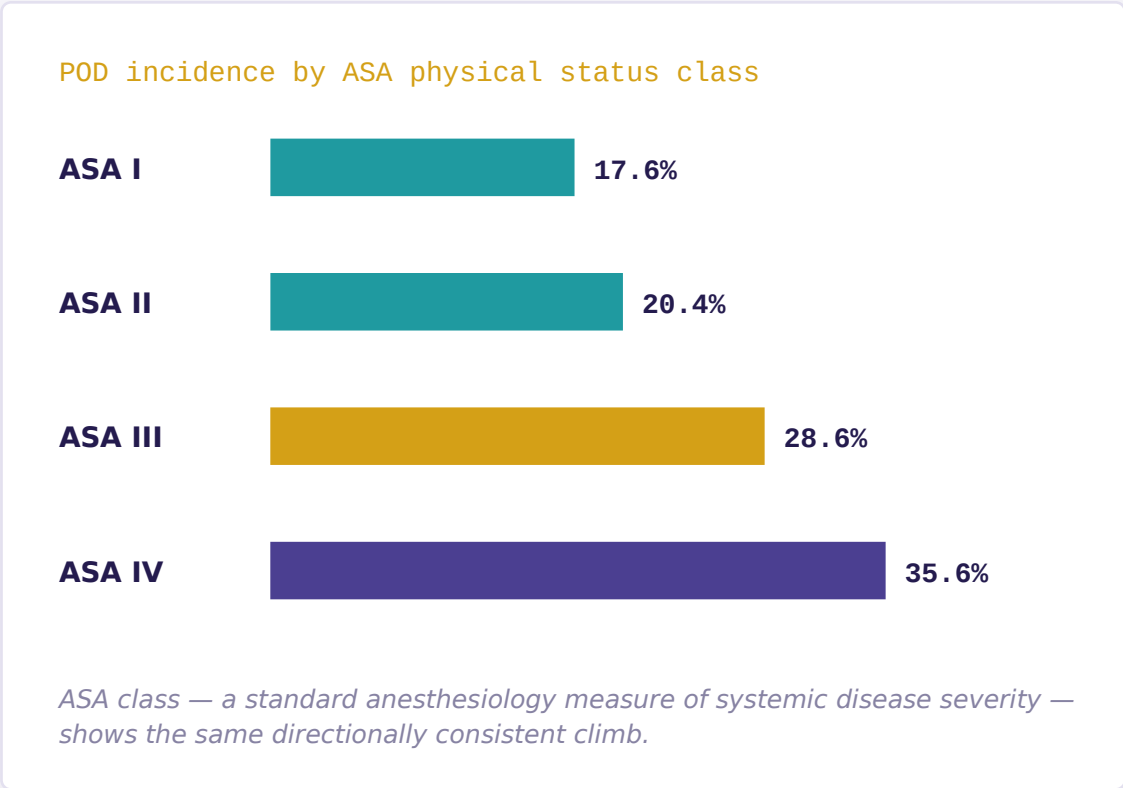
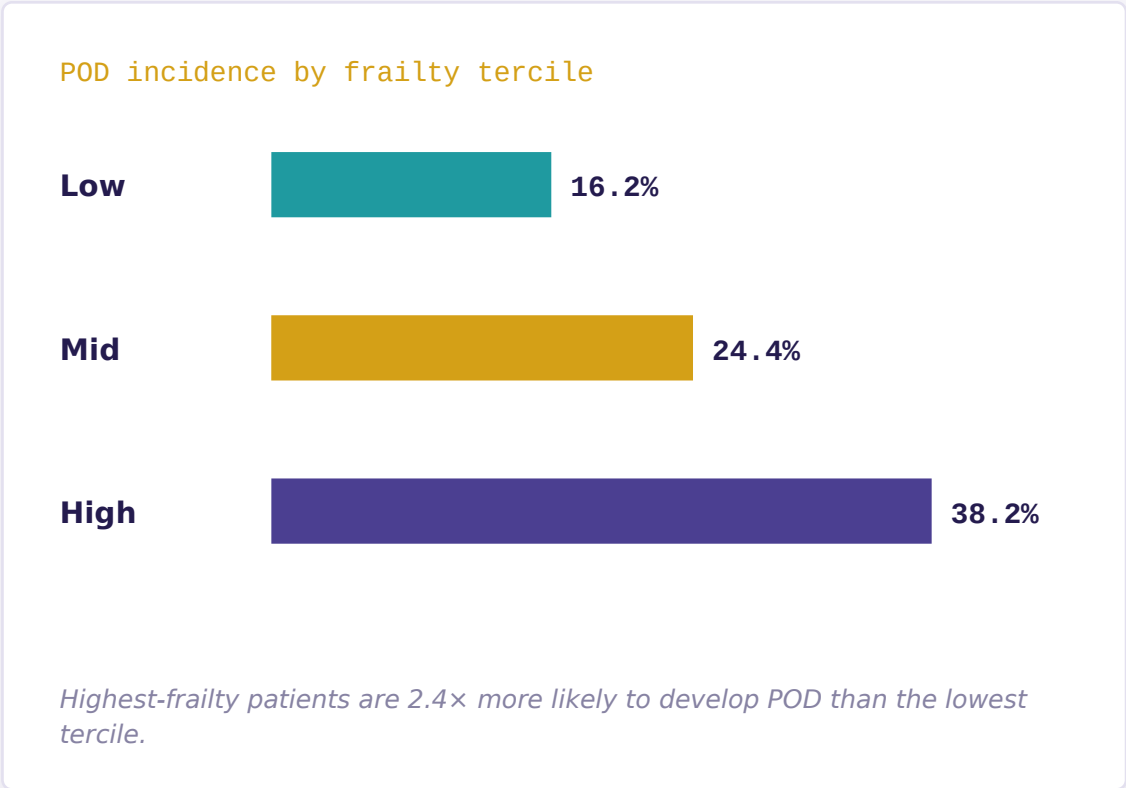
0.700

mean cross-validated AUC-ROC

THE RISK GRADIENT

Frailty and surgical risk class both track POD

Two independent, well-established clinical measures both show a real, monotonic gradient with POD incidence in this cohort — a sanity check the model needs to reproduce before it can be trusted.



Continuous movement, not a single check-in

Each episode carries a full pre-op week of wrist-worn actigraphy — continuous activity counts at high temporal resolution, summarized into five circadian-rhythm features the model actually uses.



7 days

of continuous pre-op wrist activity per episode, sampled at minute-level resolution



250

episodes with raw traces available for inspection in the demo sample



5

engineered circadian features summarizing each week's rhythm (Slide 19)



Why raw traces matter, even unused

The pre-op huddle dashboard (Slide 36) shows a patient's raw 24-hour trace alongside the score — a clinician can see the pattern behind the number, not just the number itself.

Clinical history and wearable-derived rhythm



Pre-op clinical

Age, ASA class, frailty index, baseline cognition (MMSE), anticholinergic burden (ACB score), planned surgical duration, expected blood loss, anesthesia type.



Actigraphy-derived

Amplitude, interdaily stability, intradaily variability, sleep efficiency, and night-time mean activity — five standard circadian-rhythm summary statistics.

8 clinical features + 5 actigraphy features = 13 total — deliberately excluding sex and a raw self-reported sleep-efficiency field in favor of the actigraphy-derived version (Slide 18).

THE BASE RATE PROBLEM

26.2% incidence — imbalanced, not rare

A 26.2% positive rate is imbalanced enough to matter for model training, but common enough that this isn’t a needle-in-a-haystack problem — an important distinction for how the model is built (Slide 23).

Cohort composition, n = 4,200



1,101 POD episodes and 3,099 non-POD episodes — roughly a 1:2.8 ratio, addressed with class-weighted training rather than resampling (Slide 23).

A synthetic cohort, disclosed as such



Why synthetic

A generated cohort removes any dependence on identifiable patient records for a portfolio piece, while preserving realistic clinical risk gradients and a genuine actigraphy signal.



What was preserved

Real, monotonic frailty and ASA-class gradients (Slide 11), a plausible incidence rate, and an actigraphy ablation lift consistent with published wearable-sleep research patterns.



What is disclosed

No episode here corresponds to a real patient. Every downstream number is scoped to this synthetic cohort until validated on a real clinical dataset.



A methodological choice, stated up front

No claim in this deck should be read as validated on real patient data — that validation is named explicitly as a precondition for any clinical use (Slide 38).



SECTION THREE

Feature Engineering

What the model is allowed to see — and a deliberate exclusion made for reasons beyond raw predictive power.

THE FEATURE SET

Thirteen features, chosen deliberately

The final feature list is smaller than the raw data available — several columns were left out on purpose, not by oversight.

Used by the model (13)

age	asa_class
frailty_index	baseline_mmse
acb_score	intraop_duration_min
blood_loss_ml	anesthesia_general
actg_amplitude	actg_interdaily_stability
actg_intradaily_variability	actg_sleep_efficiency
actg_night_mean_activity	

Present in data, excluded (3)

sex

a fairness-motivated exclusion, examined on Slide 18

sleep_efficiency (self-reported)

superseded by the more objective actg_sleep_efficiency

circadian_disruption_true

left out of this feature set for this model version

KEY INSIGHT

Excluding sex cost nothing — and the data confirms it

“POD incidence is 26.20% for female episodes and 26.23% for male episodes — statistically indistinguishable. Sex carries no independent signal this model would have gained by including it.”



A design choice, not a limitation

The feature was deliberately left out of training — fewer sensitive attributes in a clinical model means fewer ways it can encode unwanted bias.



Verified, not just asserted

This deck checked the claim against the actual cohort rather than assuming it — the 0.03-point gap is well within noise.



A pattern worth repeating

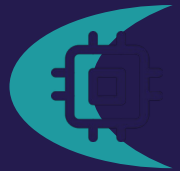
Any future feature added to this model should clear the same bar: does it change outcomes, or just correlate with a group?

Standard circadian-rhythm summary statistics

Each patient’s raw week-long activity trace is reduced to five established rest-activity rhythm metrics before the model ever sees it — the same summary statistics used in circadian-rhythm research.

Amplitude	Strength of the day/night activity rhythm
Interdaily stability	How consistent the rhythm is from day to day
Intradaily variability	How fragmented activity is within a single day
Sleep efficiency	Share of time in bed actually spent asleep
Night mean activity	Average movement during the overnight window

These five features together carry the actigraphy signal responsible for the +8.7% AUC lift in Section 05.



SECTION FOUR

Modeling: Gradient-Boosted Risk

A LightGBM classifier trained across clinical and wearable features, weighted for the 26% base rate.

One model, two feature families, one probability

Unlike some of the multi-stage pipelines in this portfolio, POD risk uses a single gradient-boosted classifier — clinical and actigraphy features are simply concatenated into one feature vector.



13 features

*8 clinical + 5 actigraphy,
concatenated into one vector per
episode*



■ LightGBM classifier

300 shallow trees,
class-weighted for the
26% base rate



P(POD)

*one calibratable probability,
thresholded for the pre-op huddle flag*

MODEL CONFIGURATION

Shallow, regularized trees on a modest dataset

4,200 episodes is small by modern ML standards — every hyperparameter here leans toward avoiding overfitting rather than squeezing out marginal AUC.

n_estimators Boosting rounds 300	learning_rate Conservative shrinkage 0.05
num_leaves Max leaves per tree 31	class_weight Reweights the minority POD class balanced
max_depth Shallow trees — controls overfitting 4	subsample Row sampling per tree 0.85
min_child_samples Minimum samples per leaf 20	colsample_bytree Column sampling per tree 0.85

Reweighting, not resampling

With 26.2% positive cases, this is a mild-to-moderate imbalance — addressed with class weights inside the loss function, not by duplicating or discarding rows.



class_weight = "balanced"

Every POD episode's training loss is scaled up in inverse proportion to its class frequency — no synthetic rows, no discarded data.



Every real episode stays real

Unlike oversampling (SMOTE-style techniques), this never invents a synthetic patient — every training example the model sees is a genuine cohort episode.

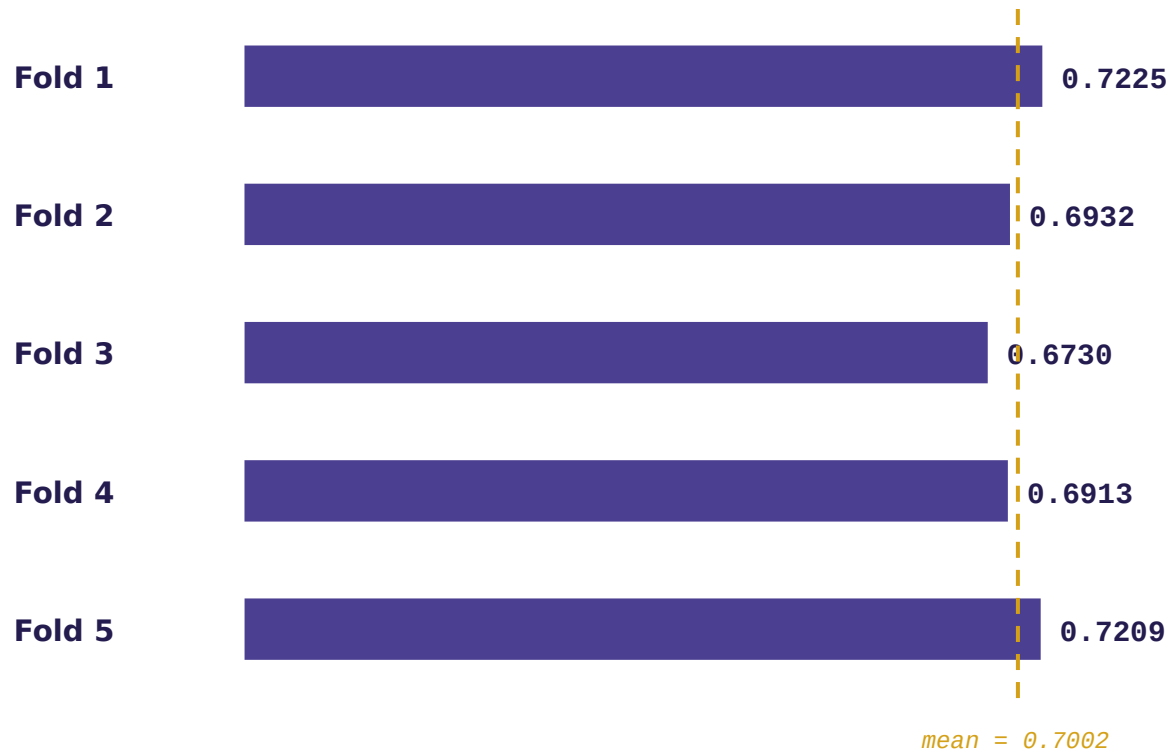


Why this matters here

At 26% positive, aggressive resampling risks manufacturing patterns that aren't really there — reweighting is the more conservative choice at this imbalance level.

Five-fold cross-validation, reported fold by fold

AUC-ROC by fold



Reading the spread

0.673 to 0.7225

fold-level range — a real but modest spread across folds

No fold collapses

every fold clears 0.67 — no single unlucky split is propping up the mean

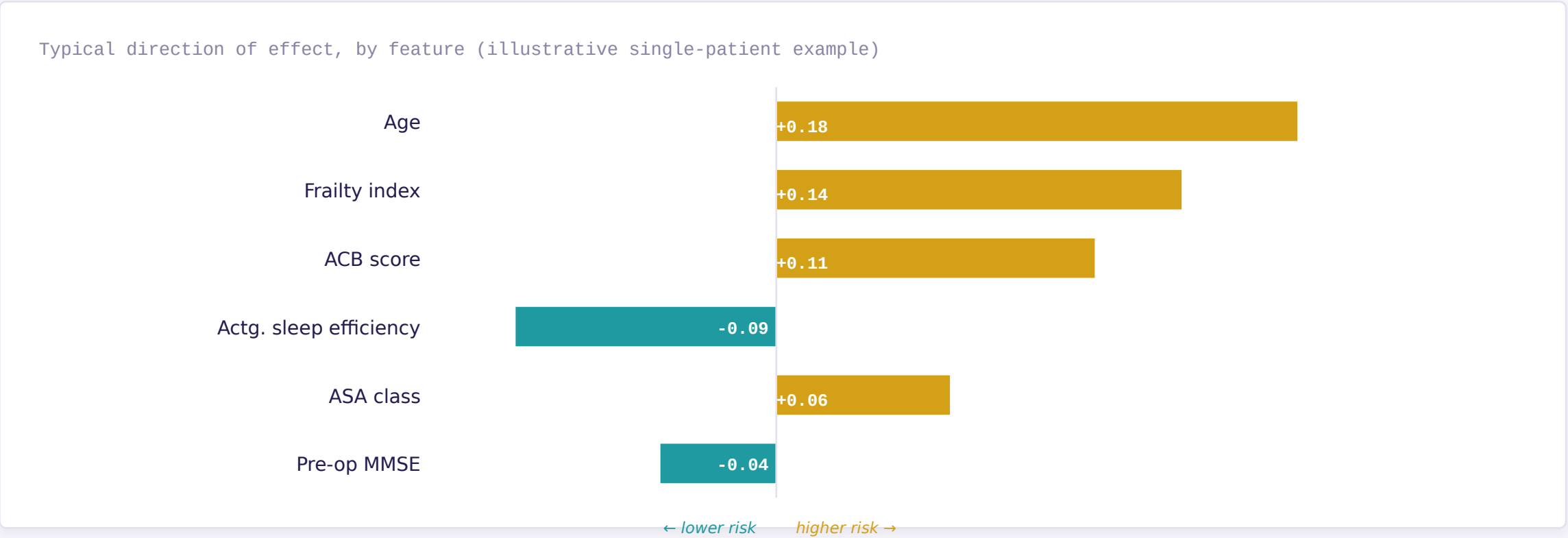
A stable, if modest, signal

the model's discrimination is consistent, not a fragile artifact of one train/test split

EXPLAINING EVERY PREDICTION

SHAP values, computed per patient, at the huddle

The dashboard doesn't just show a number — it shows which features pushed that specific patient's score up or down, using SHAP (SHapley Additive exPlanations) values computed live.





SECTION FIVE

Evaluation & Validation

Discrimination, lift from the wearable signal, and an honest look at how many cases the flag actually catches.

Out-of-fold predictions, three questions

Every evaluation number in this section comes from out-of-fold predictions — each episode scored only by a model that never trained on it, across all 4,200 episodes.



Does it discriminate?

AUC-ROC — do POD episodes score higher than non-POD episodes, across every threshold?



Does the wearable data help?

A controlled ablation — the same model, with and without actigraphy features, nothing else changed.

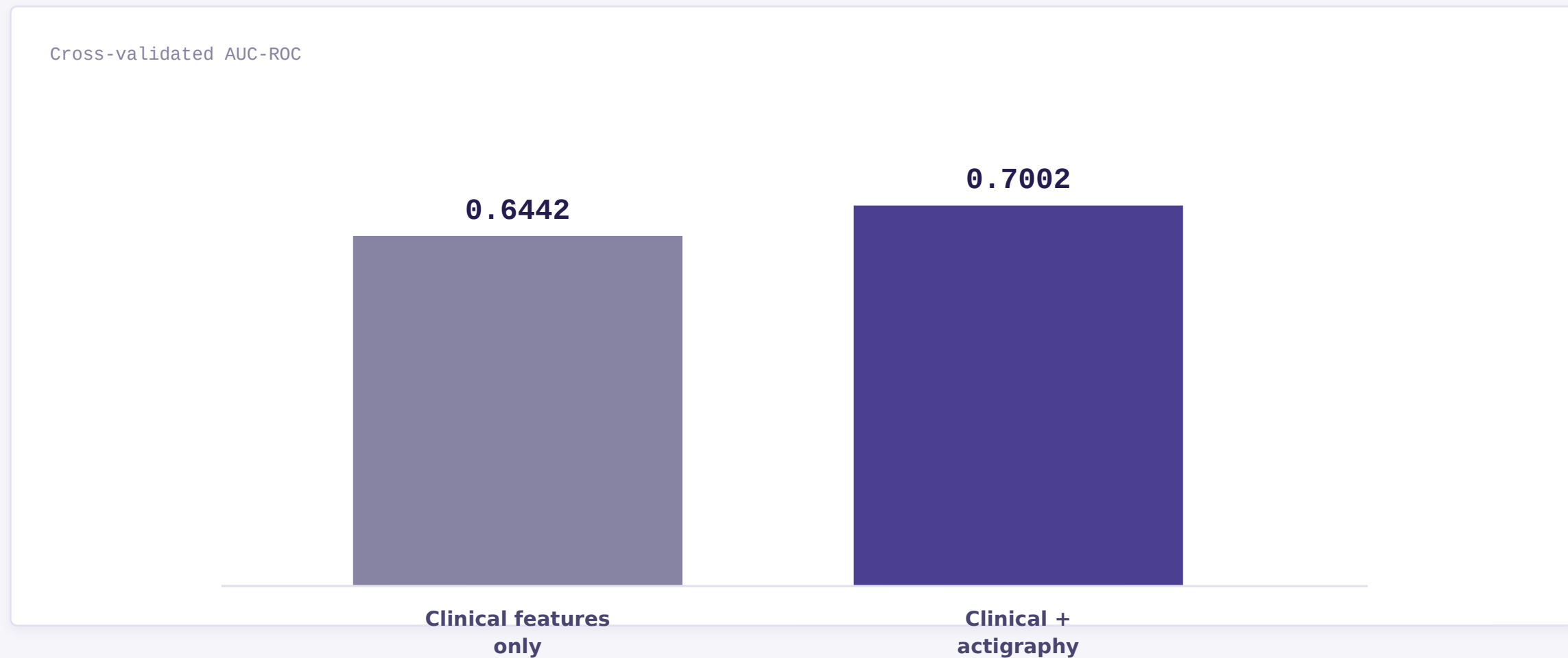


What happens at the operating point?

Precision and sensitivity at the actual flagging threshold — the numbers that matter for a real huddle.

THE HEADLINE CHART

Tabular-only vs. tabular + actigraphy



A week of sleep data earns its place in the model

“Adding five actigraphy features lifts AUC from 0.6442 to 0.7002 — an 8.69% relative gain, clearing the 8% target the ablation study was designed to test.”



A controlled comparison

Same model, same folds, same episodes — the only thing that changed between the two bars on Slide 28 is whether actigraphy features were included.



Sleep carries real signal

The lift confirms what the SHAP chart (Slide 25) already hinted at — `actg_sleep_efficiency` is a genuine, independent contributor, not noise.



Still a modest ceiling

An 8.7% relative lift is real, but it moves AUC from weak to modest — not from modest to excellent (Slide 34).

THE HONEST MISS RATE

At the operating threshold, most cases go unflagged

Precision and sensitivity pull in opposite directions — reporting one without the other would be misleading for a screening tool meant to catch real cases.

840

episodes flagged — the top 20%, at threshold 0.614

50%

precision — of those flagged, half go on to develop POD

38%

sensitivity — of all 1,101 POD cases, only 420 are flagged in advance



Why this is disclosed, not softened

A screening tool this misses 681 of 1,101 real cases is not a substitute for clinical vigilance — it is a supplementary signal, and the model card (Slide 37) says so explicitly.



SECTION SIX

Clinical Impact

What a risk flag actually changes at the pre-op huddle — and why a modest tool can still be a useful one.

WHAT A FLAG ACTUALLY TRIGGERS

A low-risk bundle, not a high-risk decision

The consequence of a false positive matters as much as the number itself — and here, the consequence is deliberately mild.



The Hospital Elder Life Program

An established, non-pharmacological bundle — reorientation, early mobility, sleep hygiene, vision/hearing support, and hydration protocols.



Broadly beneficial, low downside

These interventions carry little risk even for patients who would never have developed POD — unlike, say, a preventive medication.



Why 50% precision is tolerable here

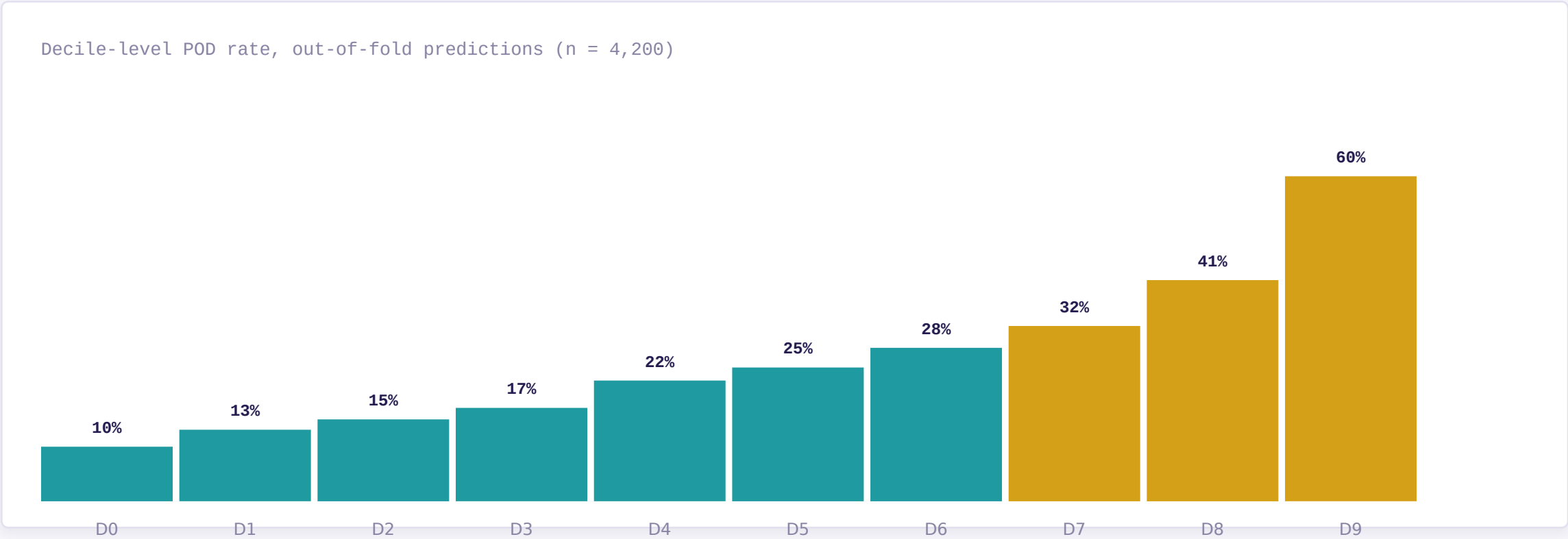
A coin-flip false-positive rate would be a serious problem for a risky intervention. For this bundle, it means extra attentive care for patients who didn't strictly need it.

This is a description of the tool's intended workflow, not medical advice — clinical protocols are set by the care team, always.

SEEING THE WHOLE COHORT AT ONCE

Where one patient sits against everyone else

The pre-op huddle dashboard (Slide 36) plots a single patient’s score against the full distribution of scores — context a bare percentage doesn’t provide on its own.



D9 (top-scored 10% of episodes) shows nearly 6× the POD rate of D0 — the model orders risk sensibly even though its miss rate (Slide 30) remains real.

A modest ceiling, stated plainly



What it tells us

The model ranks POD-bound episodes higher than non-POD episodes 70% of the time for a random pair — a real, better-than-chance signal, confirmed stable across 5 folds.



What it doesn't tell us

It says nothing about individual diagnosis, nothing about causation, and nothing about performance on a population different from this synthetic cohort.



A disclosed dashboard discrepancy

The reference dashboard displays a 28% risk threshold; the model bundle's actual saved threshold is 61.4%. This deck uses the model file, not the mockup, throughout.

An honest ceiling, stated once, is worth more than an impressive-looking number that quietly doesn't hold up under a second look.



SECTION SEVEN

Deployment & Responsible Use

A dashboard, a model card, and the guardrails a clinical screening tool needs before it earns a place in a real pre-op huddle.

The pre-op huddle dashboard, at a glance



Model card summary



Intended use

A pre-op screening aid that estimates POD risk to help direct the Hospital Elder Life Program bundle. Not a diagnostic tool; not validated on real patient data; always reviewed by a clinician.



Limitations

AUC 0.70 on a synthetic cohort; 38% sensitivity at the operating threshold means most POD cases are not flagged in advance. Requires real-world validation before any clinical use.



Next steps

Validate on a real, IRB-approved clinical cohort; recalibrate the flagging threshold with clinical stakeholders; monitor for performance drift and subgroup fairness over time (Slide 39).

Guardrails for a tool that touches patient care



A screening aid, never an autonomous decision

Every flag is reviewed by a clinician at the pre-op huddle — the model surfaces a number and its drivers; a person decides what happens next.



No deployment without real-data validation

Every result in this deck is scoped to a synthetic cohort. IRB-approved validation on real, de-identified patient data is a precondition, not a formality.



Sensitive attributes excluded by design

Sex was left out of training and the exclusion was checked against the data (Slide 18) — the same bar should apply to any feature considered in the future.



Miss rate reported alongside every win

The 38% sensitivity at the operating threshold (Slide 30) ships with the AUC and lift numbers, every time — never presented in isolation.

From reference implementation to a validated tool



Real-cohort validation

Re-train and re-evaluate on an IRB-approved clinical dataset before any pilot use — every number in this deck resets at that point.



Threshold recalibration with clinicians

Set the operating threshold jointly with the care team, weighing sensitivity against alert fatigue — not just the top-20% statistical convenience used here.



Subgroup performance monitoring

Track AUC and calibration across age, frailty, and other subgroups on an ongoing basis, not just at initial validation.



Prospective shadow-mode trial

Run the score in parallel with usual care, visible but non-actionable, before it influences a single real pre-op decision.

KEY TAKEAWAYS

A modest, honestly-reported signal — not a diagnosis



0.700 cross-validated AUC-ROC



+8.7% lift from actigraphy



38% sensitivity at threshold



Sex excluded, verified fair

Skills demonstrated

Clinical risk modeling · wearable/actigraphy feature engineering · SHAP explainability · fairness-aware feature selection · responsible clinical-AI reporting

Anthony Chinedu Echem

Machine Learning Engineer & Data Scientist



Portfolio contact available on request